

Klystron Voltage and Current Waveform Checks.

CLS Test Procedure – 1.7.32.4 Rev. 0

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1.0 PURPOSE

This document outlines the procedure required to check the Klystron Voltage and Current Waveforms.

1.1 BACKGROUND

Regular preventive maintenance can prevent catastrophic failure of components of the Modulators.

1.2 QUALIFICATIONS

Electronic Technician, Technologist or Engineer.

1.3 EQUIPMENT AND MATERIALS

Tektronix TDS30xxB Oscilloscope

710pF Video Test Cable

2Mtr. RG58 Trigger Cable

2.0 PROCEDURE FOR CHECKING THE KLYSTRON VOLTAGE AND CURRENT WAVEFORMS

Plug the Oscilloscope into a network drop to facilitate printing of the Waveforms.

Turn on the Oscilloscope.

Turn on Channel 1.

Set the Vertical to 1V/cm.

Set the Horizontal to 1uSec/cm.

Press the Vertical "Menu" button.

Press the button below "Offset".

Adjust the "Coarse Control" for approximately a -50 volt offset.

Press the button below "Coupling".

On the right hand side of the screen, press the button beside the "Omega"; 1M for the Voltage check and 50 for the Current check.

Press the button beside "DC".

Ensure that the "Probe Setup" reads "1X".

Ensure that "Position" reads "0.00 div".

Connect a 2mtr. cable between CH4 of the oscilloscope and the trigger jack on the Modulator under test.

Press the Trigger "Menu" button.

Press the button beside Ch4 on the right side of the oscilloscope screen.

In the trigger section of the oscilloscope controls, press the "Set to 50%" button.

Turn off the High Voltage on the Modulator under test.

Using the 710pF Cable specially made for viewing the Voltage and Current Waveforms (RG58, 21 feet long kept in the Modulator Room Rm. 1032), connect channel 1 of the Oscilloscope to the appropriate Jack on the Klystron Tank. The Jack directly below the Pulse Cable is the Klystron Voltage Test Jack. The Jack to the right of that is the Klystron Current Test Jack.

Turn on the Modulator High Voltage.

Obtain an accurate Waveform reading on the Oscilloscope. Press the Vertical "Menu" button.

Press the button below "Offset". Adjust the "Coarse Control" until the average of the bottom of the Waveform is on the centre line. Ensure that the bottom of the waveform is flat +or – 0.5V. If this is not the case, perform the P.F.N. Tuning Procedure 1.7.32.5.

Press the Print button in the lower left corner to print the screen.

Turn off the Modulator High Voltage.

Move the cables to the other Jack and repeat the procedure using 50 ohm termination for current waveform checks

. Continue until all Modulators have been checked both Voltage and Current and all Waveforms have been printed.

$V_{kly} = \{\text{Oscilloscope reading, terminated at 1M ohm, times divider ratio}\}$ (Divider ratio is different for each Klystron Tank i.e. Pulse Transformer). Divider ratio is approximately 5000:1.

$I_{kly} = \{\text{Oscilloscope reading, terminated at 50 ohms, times Current Transformer Volts/Amp}\}$. Volts/Amp is approximately 50.

Typical values: $V_{kly} = \sim 250 \text{ kV}$

$I_{kly} = \sim 250 \text{ A}$